# Assignment#02
# Business Process Evolution: A BPMN Study

Top 5 product companies and the evolution of two business processes each, from first version to latest

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**Scope.** Five companies that provide **products** were selected.For each company, two core business processes are modelled in BPMN 2.0 at two points in time: the earliest recognisable version and the latest version. Each diagram shows participants as lanes, tasks, events, gateways and the information artifacts (data objects) involved. A table after each pair lists the information artifacts and resources of both versions, followed by milestones.

**BPMN key:** green circle = start event; red thick circle = end event; rounded box = task; diamond = exclusive gateway (Yes/No); lane = participant / resource; dashed line to page icon = information artifact (data object) used or produced by the task.

# 1. Top 5 Companies (Products)

| # | Company | Type | Why it is big |
|---|---|---|---|
| 1 | **Tesla** | Product (electric vehicles, energy) | The world's most valuable carmaker for much of the 2020s and the company that made EVs mainstream. It stands out for selling directly to customers, building its own charging network and treating the car as updatable software. |
| 2 | **Toyota** | Product (automobiles) | One of the world's largest vehicle manufacturers by volume. Its Toyota Production System (lean, just-in-time) is studied and copied across industries and is the reference model for manufacturing process improvement. |
| 3 | **Apple** | Product (consumer electronics) | One of the world's most valuable companies. Known for tightly integrated hardware, software and services, and for a highly optimised global supply chain and retail operation. |
| 4 | **Dell Technologies** | Product (PCs, servers, IT hardware) | A leading PC and enterprise hardware maker. Its direct, build-to-order model in the 1980s-90s reshaped how computers are sold and how inventory is managed. |
| 5 | **Nike** | Product (footwear, apparel) | The largest sportswear brand in the world. It pioneered outsourced, design-led manufacturing and has shifted from wholesale to a direct-to-consumer digital model. |

## Selected business processes

| Company | Process 1 | Process 2 |
|---|---|---|
| Tesla | Order-to-Delivery (vehicle sales) | Vehicle Production (manufacturing) |
| Toyota | Vehicle Production System | New Product Development |
| Apple | Sales and Distribution | Manufacturing and Supply Chain |
| Dell Technologies | Order Fulfilment | Supply Chain and Inventory Management |
| Nike | Order-to-Delivery | Product Design and Manufacturing |

# 2. Process Evolution with BPMN Diagrams

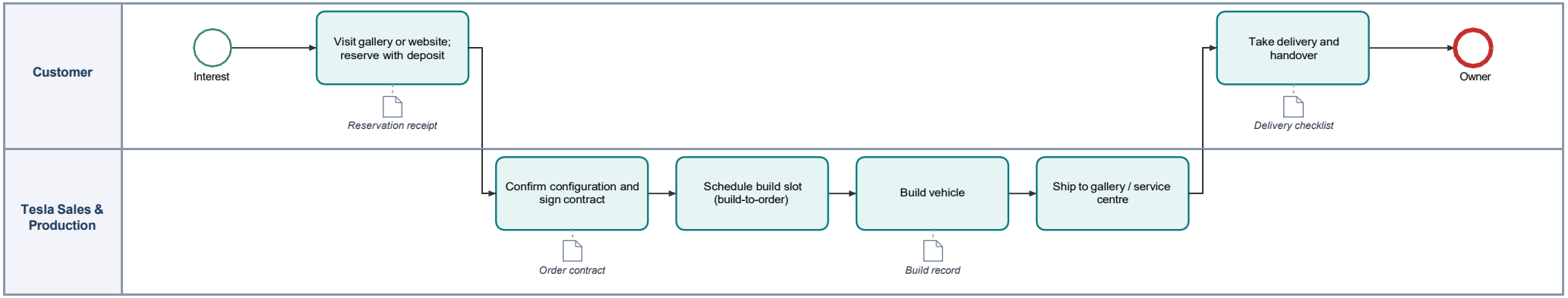
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## 2.1 Tesla

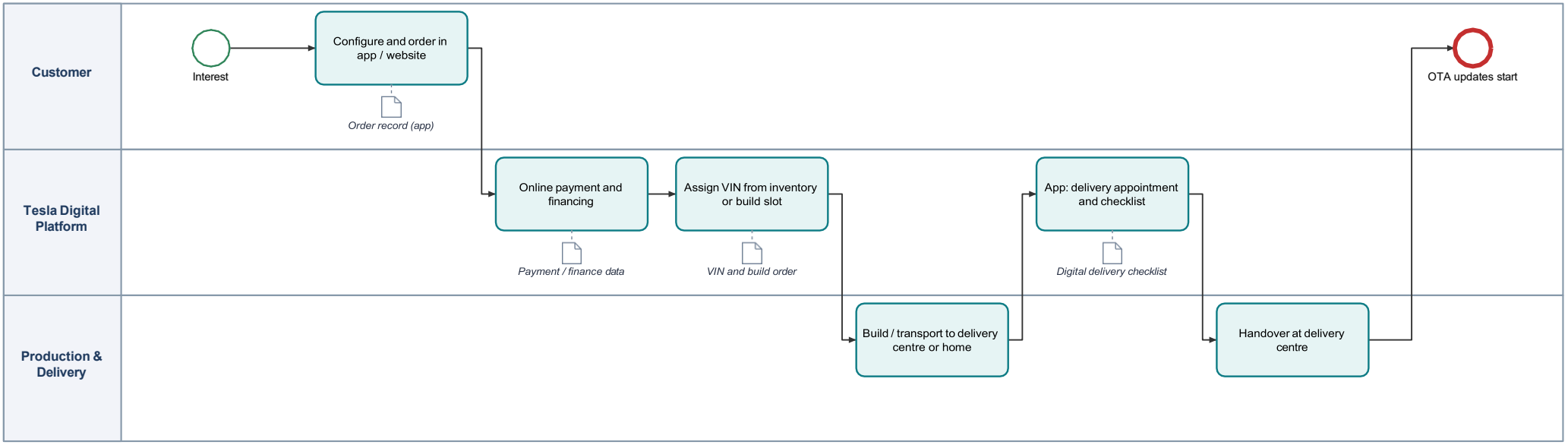
### Process 1: Order-to-Delivery (vehicle sales)

**Why this process:** Shows the move from showroom-and-deposit selling to an online, app-driven direct sales process.

**Version 1 - Early era (Roadster / Model S launch, c. 2008-2013): gallery + reservation + build-to-order**




**Latest version - Digital direct sales (c. 2019 onward): online order, app-managed delivery, OTA updates**



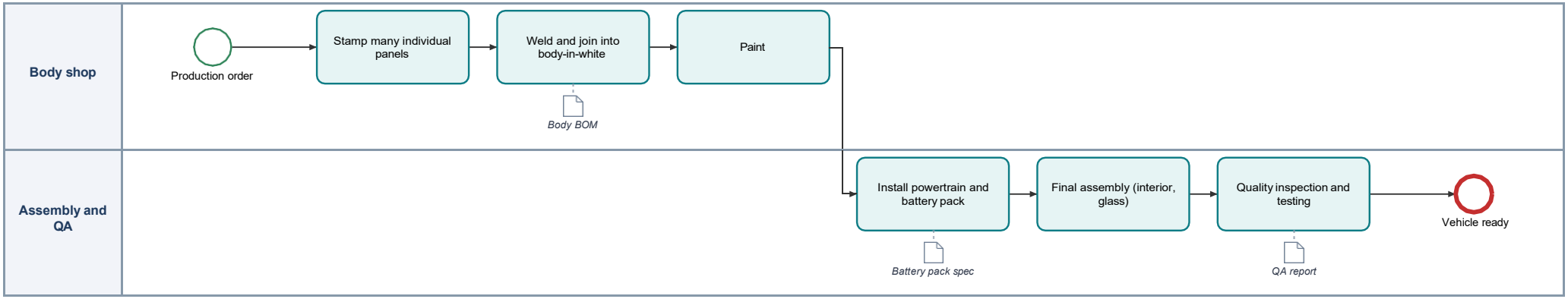
| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Reservation receipt, order contract, build record, paper delivery checklist. | Sales staff in galleries, build planners, service centres, low-volume line. |
| Latest | Digital order record, payment and finance data, VIN/build order, app checklist, OTA software package. | Tesla app and website, order-management software, delivery centres, Gigafactories, over-the-air update system. |

**Evolution milestones:** 2008: Roadster sold through galleries with deposits and build-to-order. | 2012: Model S launch; customer-specific orders and service-centre handover grow. | 2019: Tesla moves to an online-first ordering model (later adjusted). | 2020s: App-based delivery, inventory-based sales and continuous over-the-air updates.

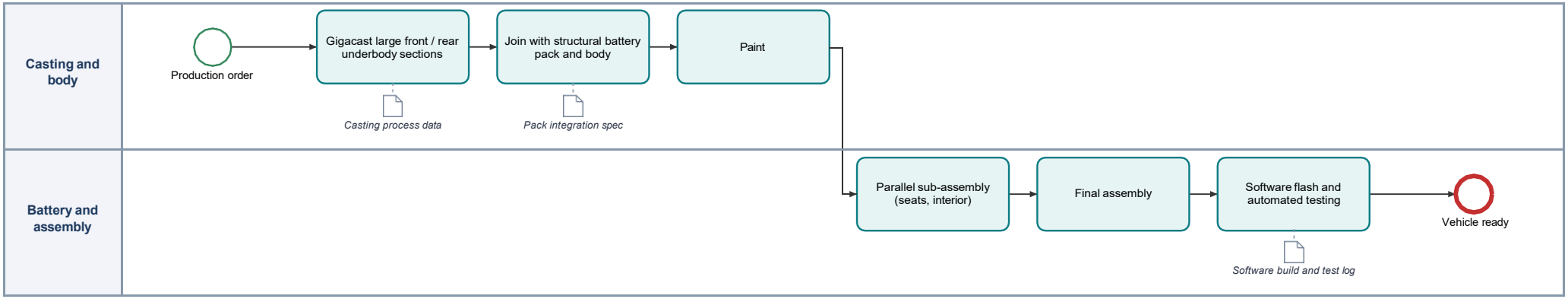
## Process 2: Vehicle Production (manufacturing)

**Why this process:** Shows the shift from many stamped and welded parts to large single-piece castings and integrated battery structures.

### Version 1 - Conventional line (Fremont Model S, early 2010s): stamp, weld, paint, assemble




### Latest version - Gigacasting and integrated structure (Model Y onward): fewer parts, parallel sub-assembly




| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Body BOM, battery pack spec, paint schedule, QA report. | Stamping presses, welding robots, paint shop, manual assembly lines. |
| Latest | Casting process data, integrated structure spec, software build and test log, line telemetry. | Giga Press casting machines, automated testing, Gigafactories (Nevada, Shanghai, Berlin, Texas). |

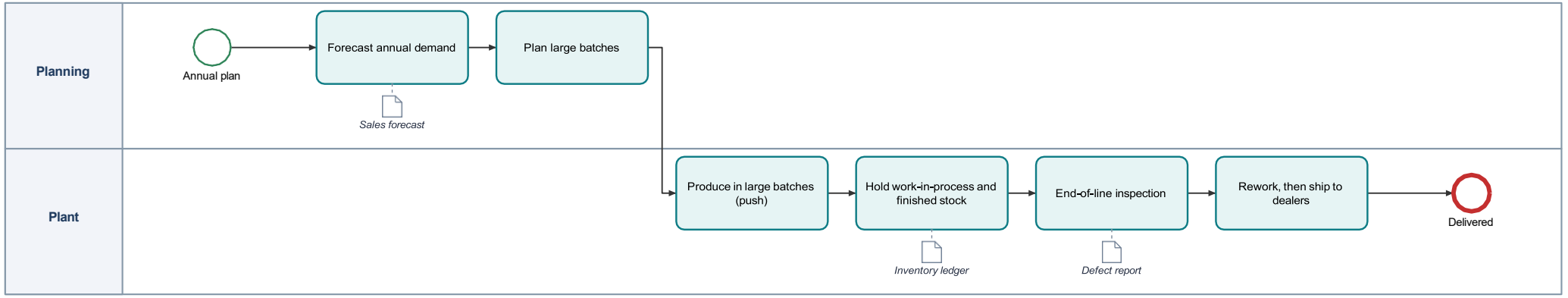
**Evolution milestones:** 2012: Model S production at Fremont (conventional stamping and welding). | 2016: Gigafactory Nevada begins battery production. | 2020: Large single-piece castings introduced for Model Y. | 2022: Berlin and Texas factories apply the newer body and battery integration.

# 2.2 Toyota

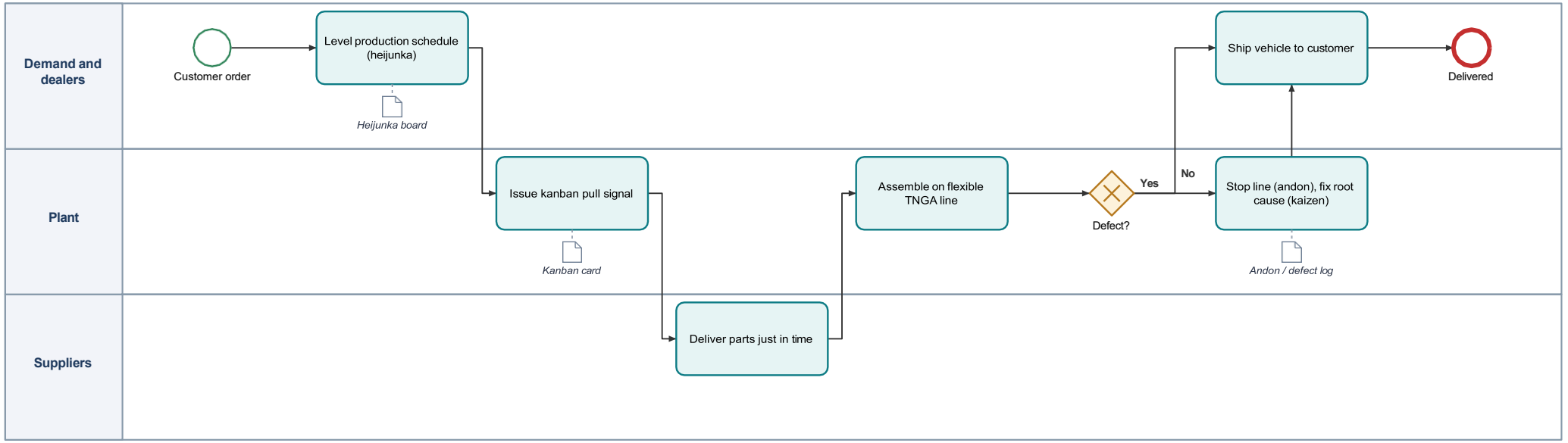
## Process 1: Vehicle Production System

**Why this process:** Shows the move from forecast-driven batch production to the pull-based, defect-stopping Toyota Production System and TNGA platforms.

### Version 1 - Batch / mass production (pre-1950s style): forecast-driven, push, inspect at the end




### Latest version - TPS + TNGA: levelled, pull-based, stop-the-line quality



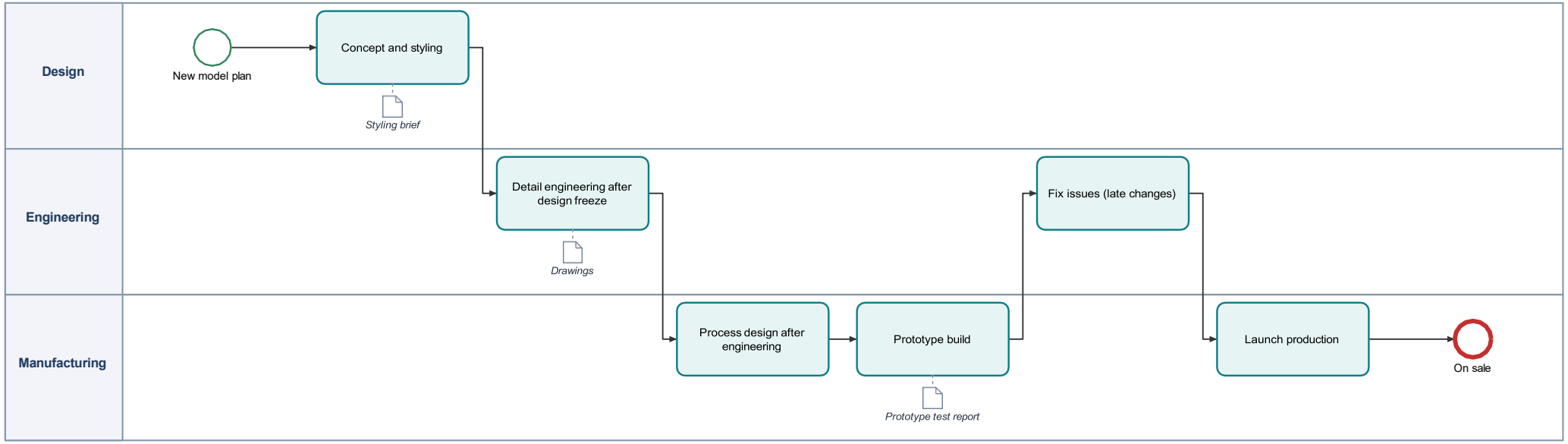
| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Annual sales forecast, batch plan, inventory ledger, end-of-line defect report. | Large stock buffers, central planners, inspection team at line end. |
| Latest | Heijunka board, kanban cards, andon/defect log, standard work sheets, platform parts catalogue. | Pull-based supply network, flexible TNGA lines, trained line workers empowered to stop the line. |

**Evolution milestones:** 1920s: Jidoka (automatic stop on defect) originates in Toyoda automatic looms. | 1950s: Taiichi Ohno builds just-in-time and kanban into production. | 1970s: TPS becomes the benchmark for lean manufacturing worldwide. | 2015: TNGA modular architecture first applied (4th-generation Prius).

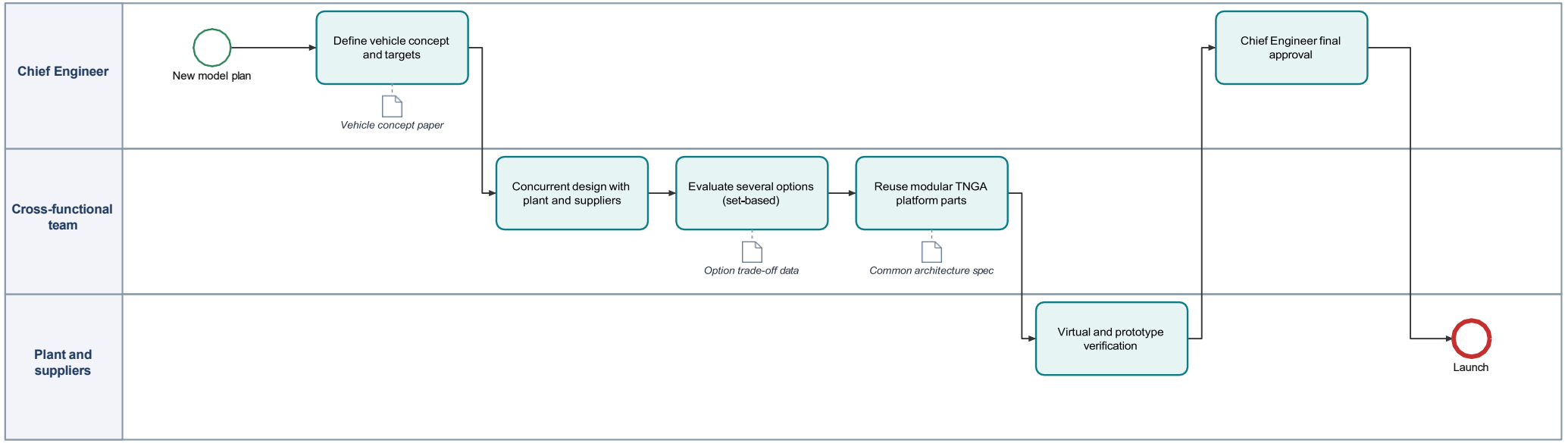
## Process 2: New Product Development

**Why this process:** Shows the shift from sequential departmental hand-offs to a Chief Engineer-led, concurrent, platform-based approach.

**Version 1 - Sequential hand-off development: design freeze first, problems discovered late**



### Latest version - Chief Engineer system, concurrent engineering and TNGA platform reuse




| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Styling brief, drawings, prototype test reports, change orders. | Separate design, engineering and manufacturing departments; physical prototypes. |
| Latest | Concept paper, option trade-off data, common architecture spec, virtual test results. | Chief Engineer, cross-functional teams, CAD/simulation tools, TNGA parts library. |

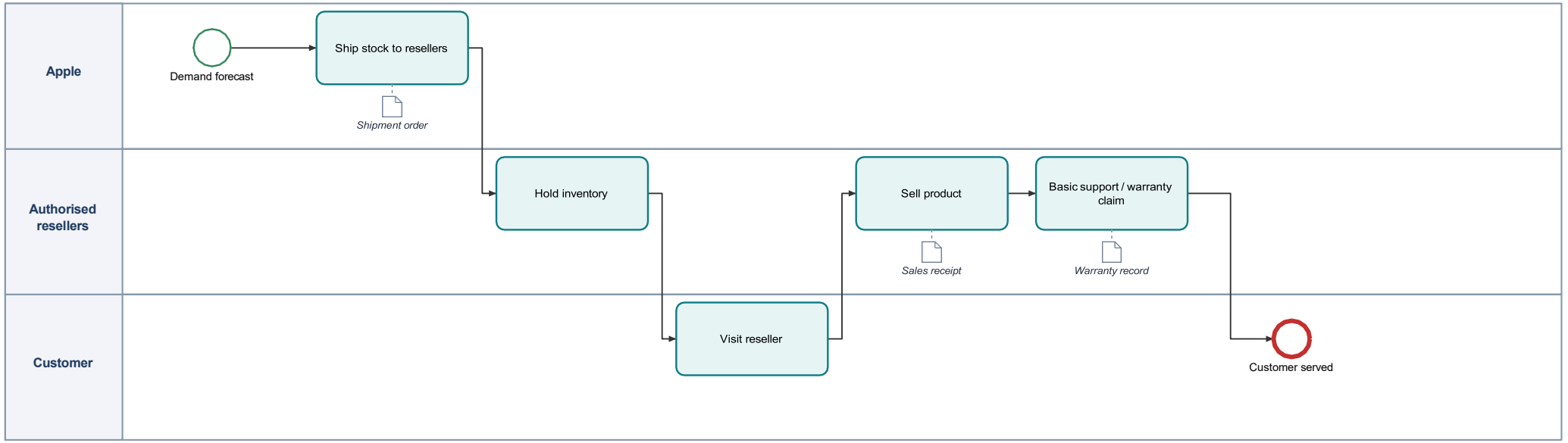
**Evolution milestones:** Mid-20th century: Departmental, sequential development is typical in the auto industry. | 1980s-90s: Toyota formalises the Chief Engineer (shusa) system and concurrent engineering. | 2012: Toyota announces the TNGA architecture-based approach. | 2015 onward: Platform reuse across models and regions shortens lead time.

# 2.3 Apple

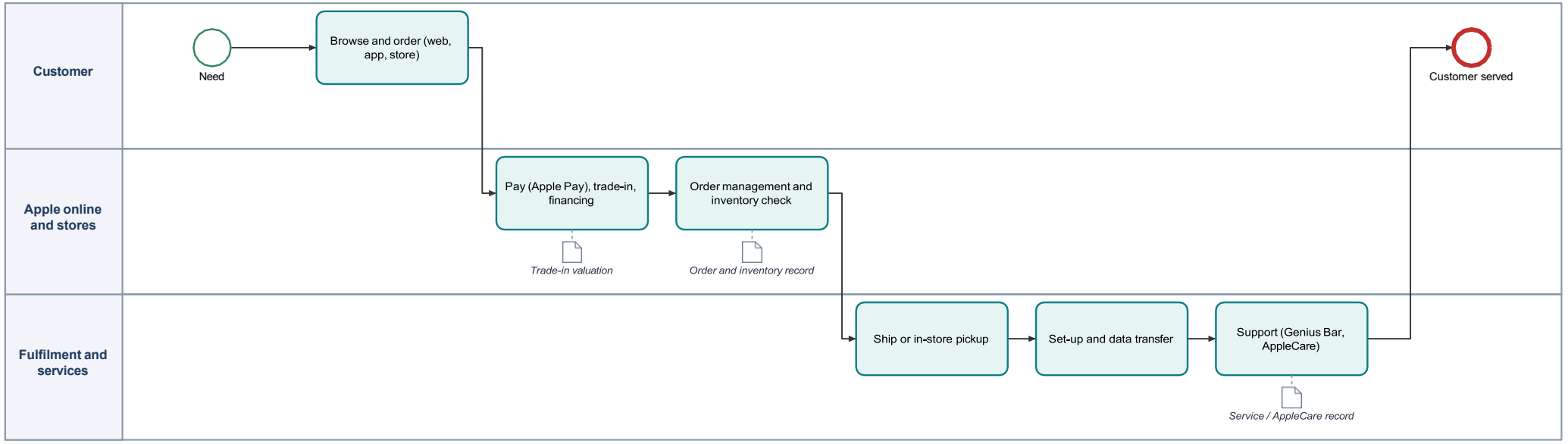
## Process 1: Sales and Distribution

**Why this process:** Shows the shift from selling through resellers to an omnichannel model of web, app and Apple Stores.

### Version 1 - Reseller-led distribution (1980s - mid 1990s)




### Latest version - Omnichannel: web, app and Apple Stores with trade-in and services



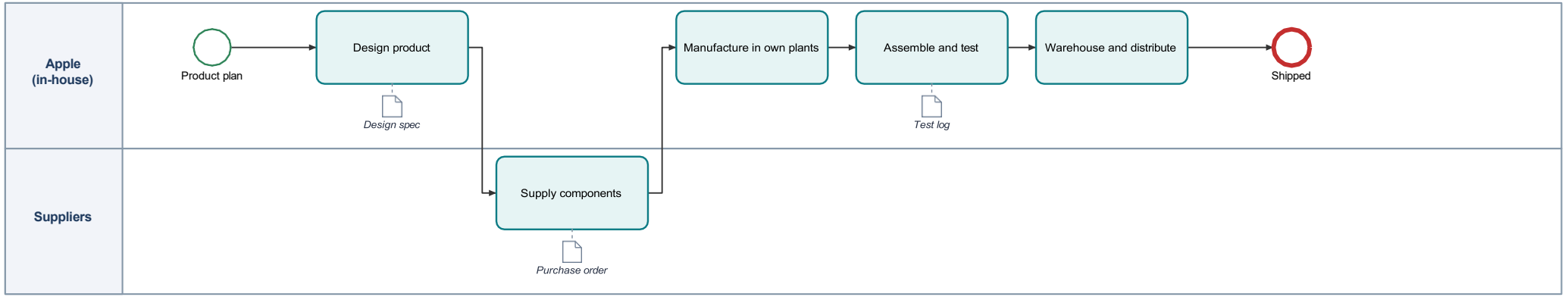
| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Shipment orders, reseller stock list, sales receipt, warranty record. | Distributors, third-party reseller staff, Apple sales team. |
| Latest | Order and inventory record, trade-in valuation, payment token, service record, customer account. | Apple Online Store and Apple Store app, Apple retail stores, fulfilment centres, Apple Pay, AppleCare. |

**Evolution milestones:** 1997: Apple Online Store launches with build-to-order Macs. | 2001: First Apple retail stores open. | 2010: Apple Store app launches. | 2014 onward: Apple Pay, trade-in and in-store pickup integrate the channels.

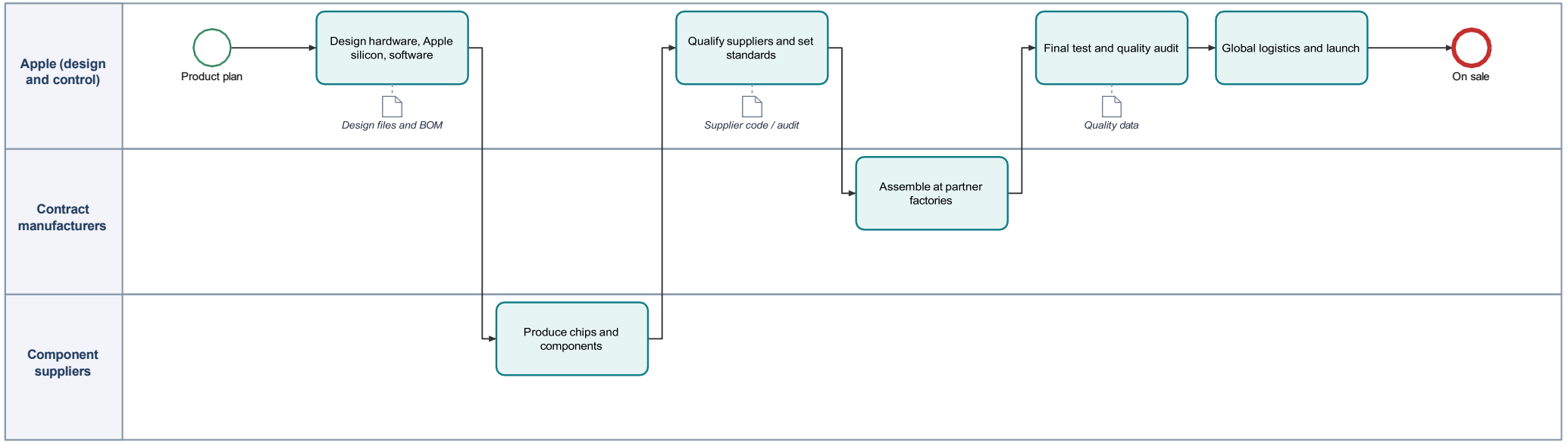
## Process 2: Manufacturing and Supply Chain

**Why this process:** Shows the move from in-house factories to a managed network of contract manufacturers and component suppliers.

### Version 1 - In-house manufacturing (1980s - mid 1990s)




### Latest version - Managed global network (contract manufacturers, in-house silicon design, regional diversification)



| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Design spec, purchase orders, test log, inventory ledger. | Own factories and staff, a smaller supplier base. |
| Latest | Design files and BOM, supplier audit records, quality data, logistics and demand plans. | Contract manufacturers in several countries, chip foundry partners, Apple operations and quality teams. |

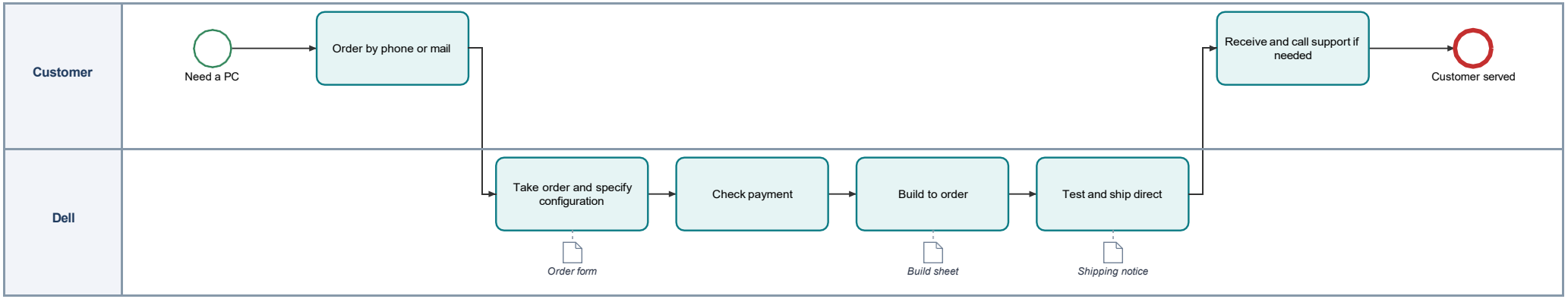
**Evolution milestones:** Mid 1980s-90s: Apple builds Macs in its own plants. | Late 1990s: Apple closes most factories, outsources assembly and consolidates suppliers. | 2020: Apple silicon begins replacing third-party Mac processors. | 2020s: Assembly spreads beyond a single country (e.g. India, Vietnam).

# 2.4 Dell Technologies

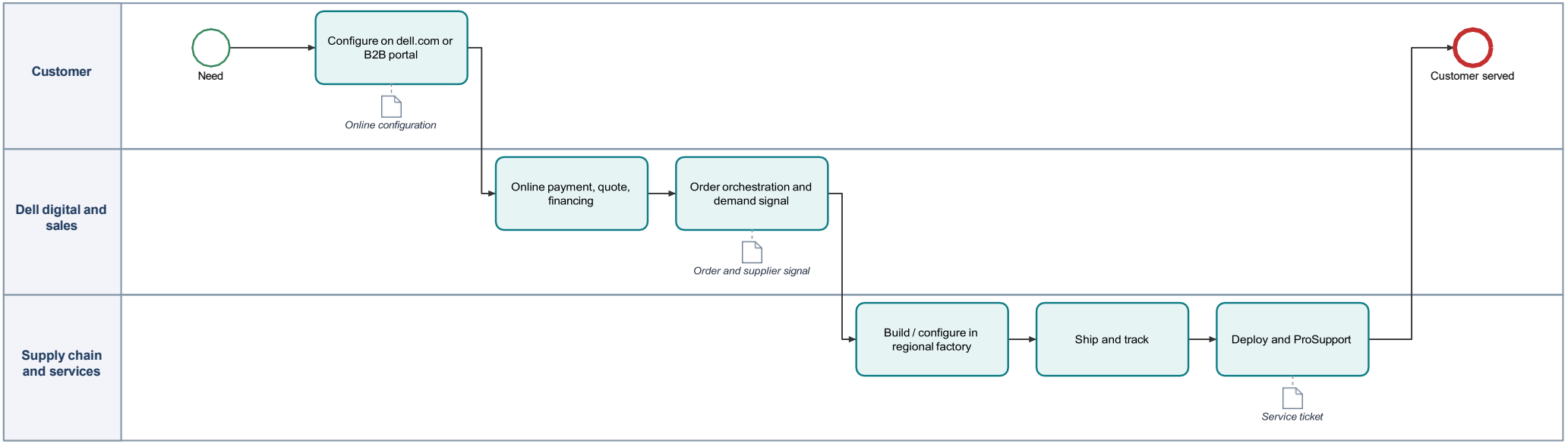
## Process 1: Order Fulfilment

**Why this process:** Shows the evolution of the direct, build-to-order model from phone and mail orders to online configuration and enterprise portals.

### Version 1 - Direct model by phone / mail (1984 - mid 1990s)




### Latest version - Online and enterprise: dell.com configuration, B2B portals, orchestrated fulfilment



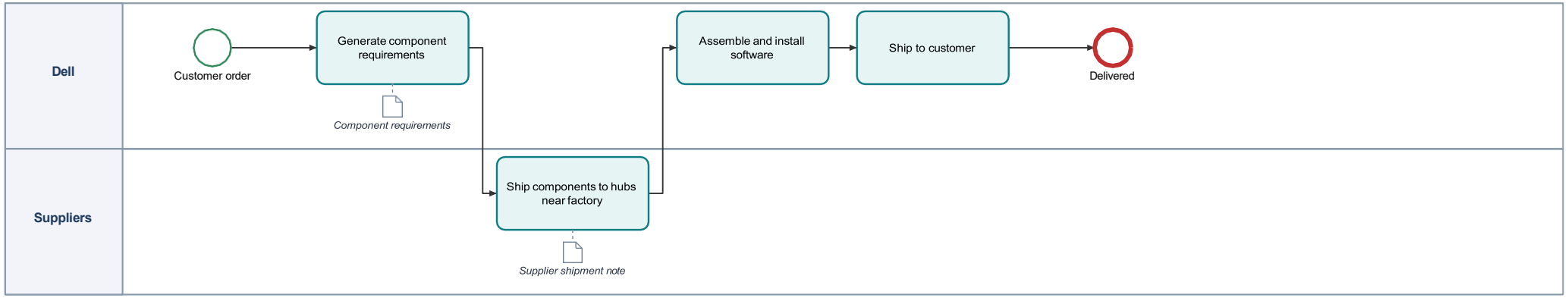

| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Paper or phone order form, build sheet, shipping notice. | Phone sales staff, assembly team, courier. |
| Latest | Online configuration, quote, order and supplier signal, tracking data, service ticket. | dell.com, enterprise portals, regional factories, logistics partners, ProSupport. |

**Evolution milestones:** 1984: Michael Dell starts selling PCs directly. | 1996: Dell begins selling online at dell.com. | 2000s: Premier / enterprise portals for business customers. | 2016: Dell Technologies formed after the EMC merger, broadening enterprise fulfilment.

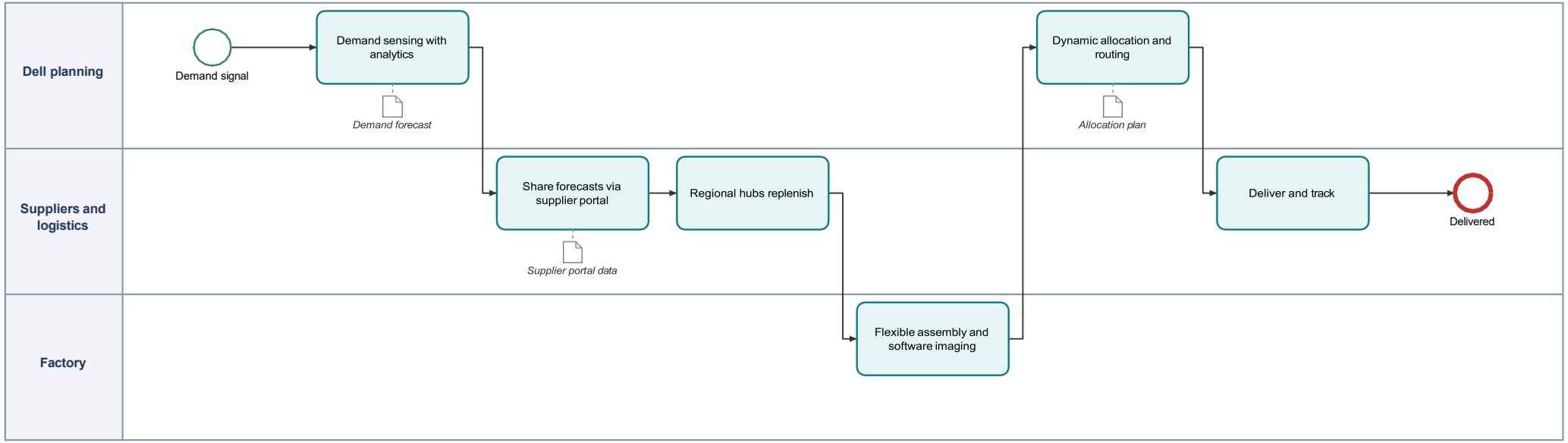
## Process 2: Supply Chain and Inventory Management

**Why this process:** Shows the evolution of Dell's just-in-time supplier model into data-driven planning and a regional hub network.

### Version 1 - Direct just-in-time model: components triggered by customer orders




### Latest version - Analytics-driven planning with regional hubs and supplier portals




| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Component requirement list, supplier shipment note, assembly record. | Dell planners, supplier hubs, assembly line. |
| Latest | Demand forecast, supplier portal data, allocation plan, tracking data. | Forecasting and analytics tools, supplier portal, regional hubs, flexible factories. |

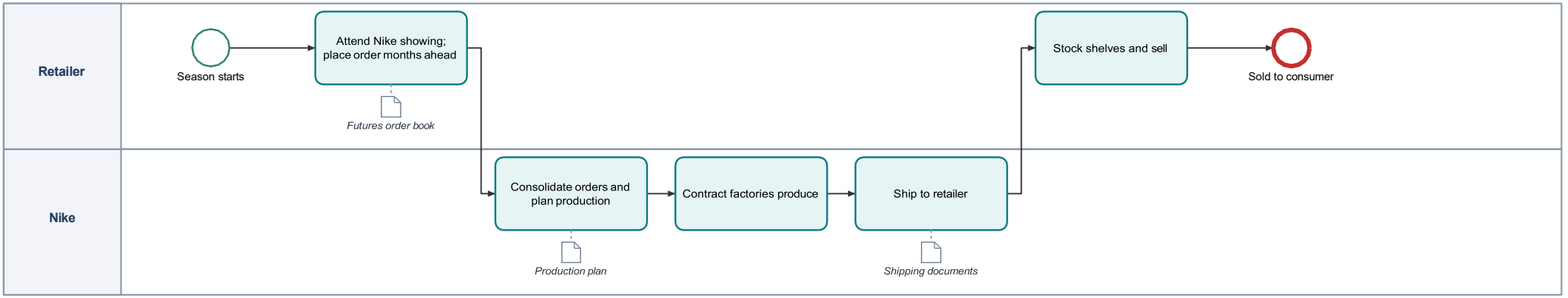
**Evolution milestones:** 1980s-90s: Direct model with very low inventory. | 2000s: Regional manufacturing and supplier hubs. | 2010s: Analytics-based demand forecasting and supplier collaboration. | 2020s: Resilience programmes after global supply disruptions.

# 2.5 Nike

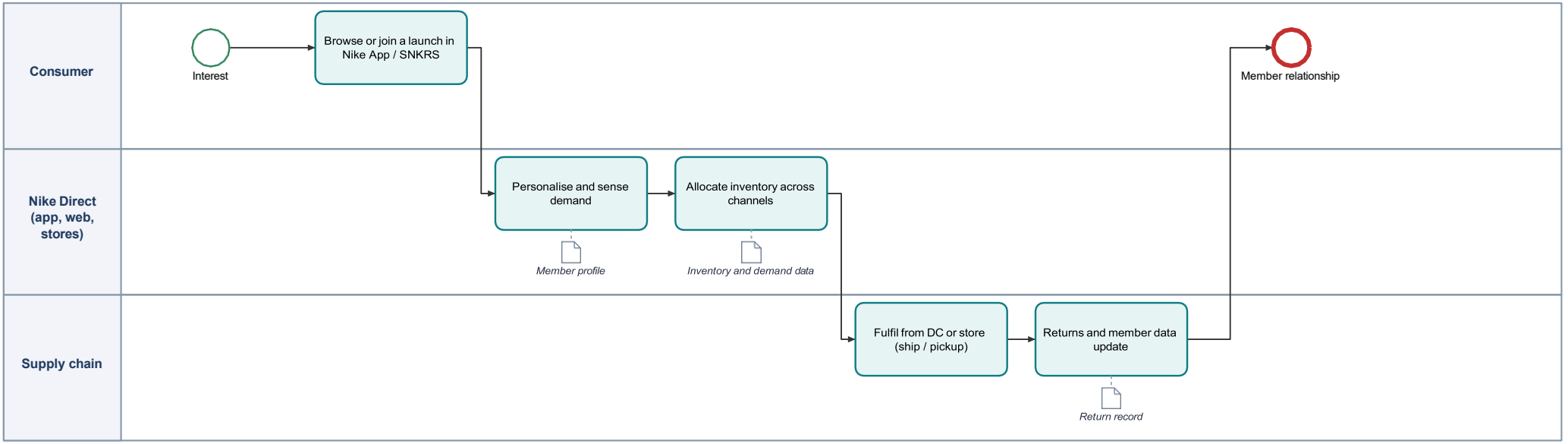
## Process 1: Order-to-Delivery

**Why this process:** Shows the shift from retailer 'futures' ordering through wholesale to a direct, app-based consumer channel.

### Version 1 - Wholesale 'futures' ordering (1970s onward)




### Latest version - Consumer Direct: Nike App / SNKRS / stores with demand sensing



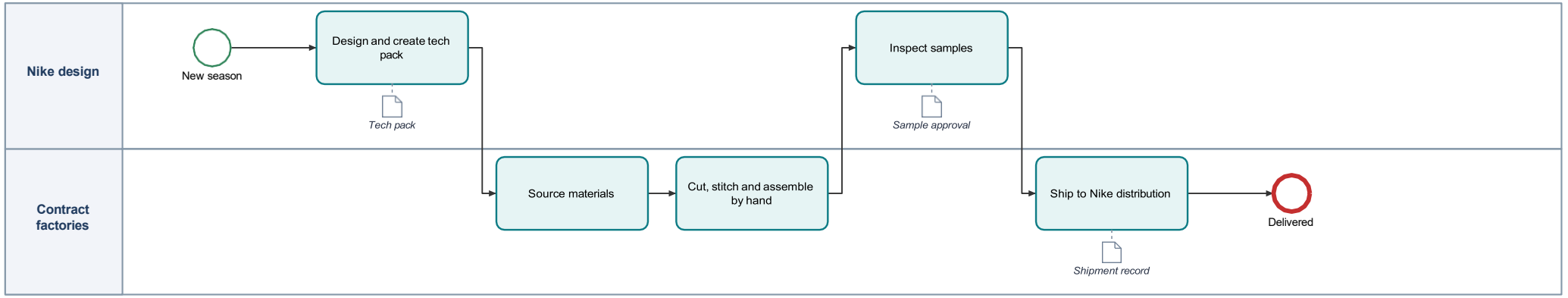

| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Futures order book, production plan, shipping documents, retailer invoices. | Account managers, retail buyers, contract factories, wholesale distributors. |
| Latest | Member profile, digital order, inventory and demand data, return record, launch allocation data. | Nike App, SNKRS, Nike.com, Nike-owned stores, distribution centres, data and analytics teams. |

**Evolution milestones:** 1970s: Futures programme lets retailers order months ahead. | 2000s: Nike.com and Nike-owned retail grow. | 2015: SNKRS app launches for limited releases. | 2017-2020: Consumer Direct Offense then Consumer Direct Acceleration shift focus to direct channels.

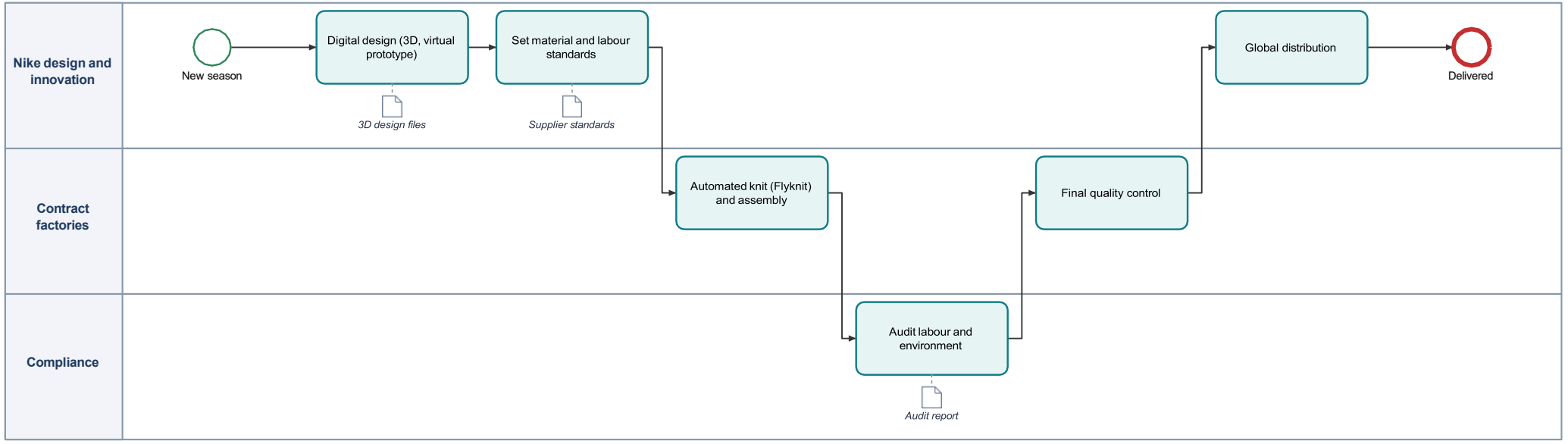
## Process 2: Product Design and Manufacturing

**Why this process:** Shows how Nike moved from hand-assembled shoes in outsourced factories to digital design, automated knitting and audited manufacturing.

### Version 1 - Design in-house, outsource hand-assembly (1960s - 1990s)




### Latest version - Digital design, automated knit / assembly and compliance audits




| Version | Information artifacts | Resources |
|---|---|---|
| Version 1 | Tech pack, sample approval, shipment record. | In-house designers, contract factories, manual assemblers. |
| Latest | 3D design files, supplier standards, audit report, quality data, distribution plan. | Digital design tools, Flyknit and automated machinery, audit teams, global distribution centres. |

**Evolution milestones:** 1960s-70s: Nike outsources production to contract factories in Asia. | 1990s: Public scrutiny of factory conditions leads to supplier standards. | 2012: Flyknit introduces knitted uppers and reduced waste. | 2020s: Digital design and compliance auditing are integral to the process.

# 3. Review and Comparison

## 3.1 Cross-company comparison (my five companies)

| Company | Process | Version 1 characteristic | Latest characteristic | Main driver of change |
|---|---|---|---|---|
| Tesla | Order-to-delivery | Gallery, deposit, build-to-order | Online order, app delivery, OTA | Cost of dealers, digital customer experience |
| Tesla | Production | Stamped and welded many parts | Gigacast, integrated structure | Fewer parts, lower cost, speed |
| Toyota | Production system | Forecast batch, push, end inspection | Pull (kanban), levelled, stop-the-line | Waste and defect reduction |
| Toyota | Product development | Sequential hand-offs | Chief Engineer, concurrent, TNGA | Shorter lead time, reuse |
| Apple | Sales and distribution | Reseller-led | Omnichannel with trade-in | Control of customer experience |
| Apple | Manufacturing | In-house plants | Managed global partner network | Scale, cost, flexibility |
| Dell | Order fulfilment | Phone/mail direct | Online, B2B portals, orchestration | Internet, enterprise customers |
| Dell | Supply chain | JIT from orders | Analytics-driven planning, hubs | Resilience and forecasting |
| Nike | Order-to-delivery | Wholesale futures | Direct app / SNKRS, demand sensing | Customer data and margin |
| Nike | Design and manufacturing | Hand-assembly, outsourced | Digital design, automation, audits | Speed, sustainability, compliance |

**Outcomes across the five companies:** (1) processes moved from **push / forecast-based** to **pull / demand-driven**; (2) manual and paper artifacts were replaced by **digital data objects** (orders, profiles, telemetry); (3) the customer moved from a passive receiver to an **active participant** in the process (app, configurator); (4) quality and compliance moved **into** the process (jidoka, audits, automated testing) instead of end-of-line inspection; (5) the number of handoffs between lanes generally fell, while the number of supporting systems rose.

# 4. References and Sources to Verify


- Tesla: company investor relations and annual reports (10-K); Tesla Impact Report; public statements on direct sales and Giga Press casting.
- Toyota: Toyota Motor Corporation 'Toyota Production System' and 'TNGA' pages; Liker, J. (2004), The Toyota Way; Ohno, T. (1988), Toyota Production System.
- Apple: Apple Newsroom (Apple Store history, Apple silicon); Apple Supplier Responsibility reports.
- Dell: Dell Technologies company history; Magretta, J. (1998), 'The Power of Virtual Integration', Harvard Business Review.
- Nike: Nike Inc. 10-K filings and Impact Reports; Nike News (Flyknit, SNKRS, Consumer Direct Acceleration).
- Notation: OMG, Business Process Model and Notation (BPMN) 2.0 specification.

